# SoK: Agent Communication Side Channels Feedback Paths and Executed Cover

***Abstract*—Classical traffic analysis explains how encrypted communication reveals activity. Which conditions carry over when an Agent forwards tool results to a model and executes privacy decoys as application work? We connect twenty-two mechanism anchors to fifteen direct Agent studies and forty-four settings through a transfer matrix, deployment access map and evidence-availability summary. The synthesis identifies two execution boundaries for protection. Feedback crosses separately controlled services, so a tool-side intervention must be evaluated against the model and recipient views that remain. Executed cover crosses into application work, so its privacy result needs an original-task constraint as well as a traffic budget. These extend established evaluation principles to the Agent execution path. Among four query/workflow privacy settings from two sources, protected-condition attack adaptation is located in two and matched original-task utility in one; none documents both in the retained records. The executed-decoy utility gap is supported by a published study; several model-view and adaptation measurements depend on preprints. A controlled feedback pilot supplies phase and complete-execution comparisons, with no multiplicity-adjusted primary rejection. The resulting design priorities are protection across services and privacy–utility measurement under the same intervention.**

## 1. Introduction

An Agent receives the instruction *process the next support ticket*. It reads a customer's request, checks an order and approval, consults a model, and submits a refund. Encryption hides messages from a network observer but leaves contacted services, traffic sizes and timing visible. The tool's answer also becomes input to another service: the model. Adding a decoy request has a second consequence if the Agent executes it: the defense now changes the work done for the user. Figure 1 follows the ticket through these services.

**The unit of protection is the execution across services.** Classical traffic analysis provides observer models, segmentation methods and adaptive evaluation. We ask which conditions these results need when applied to Agent execution. Twenty-two mechanism anchors and fifteen direct Agent studies supply two complementary layers: the established relation and the evidence for its Agent application. Forty-four setting records connect them to the interfaces in Figure 1.

Feedback connects separately protected services. Research queries can be recovered from tool destinations, and task labels can be recognized on model connections [2], [9], [4]. A contacted tool has additional access to its arguments. These observations put protections at different boundaries: tool-response padding leaves the model's later request outside its scope, while transit encryption leaves the recipient's arguments visible. Our map links each inference to its access rights and execution stage. The resulting design question is which representations remain exposed after a defense, including initial instructions, forwarded feedback and subsequent actions.

Executed cover needs an application constraint. Cover discarded at a transport endpoint consumes bandwidth; a decoy executed by an Agent also consumes application resources and may enter later decisions. In the research-Agent study WRA, five decoys perform additional research and lower query recovery, but the reported task-utility scores belong to a separate visibility intervention [2]. In the live Oblivious HTTP (OHTTP) study, adapted inference is paired with bytes and delay [3]. Reading these settings together identifies the missing measurement: the original task's outcome under the same intervention that produces the privacy result. This distinction supports a design space organized by where cover terminates and which application effects it permits.

Website fingerprinting (WF) and machine-learning security establish realistic populations, segmentation, transfer and adaptive evaluation [15], [48], [34], [16], [55], [19]. We organize their application around three contributions. A *transfer matrix* connects each established condition to the Agent setting that preserves it, adds an obligation or leaves it untested. The *deployment and defense maps* expose the two execution boundaries above and the rights needed to observe them. An *evidence-availability summary* uses question-specific denominators: four query/workflow privacy settings contain two located adaptation results and one matched original-task utility result, with no setting containing both. Identity obfuscation and adversarial-Agent blocking retain their separate objectives.

A paired mechanism study complements the synthesis with identical initial messages, category-bearing tool feedback and randomized task composition over LangGraph, Model Context Protocol (MCP) Streamable HTTP and streaming HTTP/2. Its 3,120 local executions and 192 hosted-model attempts separate first-feedback recognition, complete-execution exposure and task discovery. All twelve adjusted primary comparisons exceed .05, including post-feedback versus prefix. The pilot supplies bounded measurements; the transfer matrix states which design conclusions

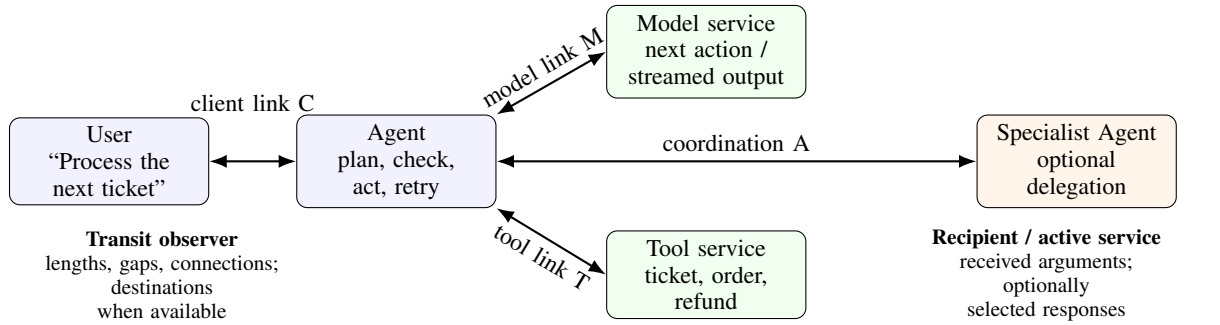



Figure 1. Running support-ticket example and observation rights. A tool result returns to the Agent and can enter the next model request. Each encrypted link exposes its own traffic shape; a recipient additionally sees the request addressed to it. Concurrent tickets may share a connection on either link. The arrows represent application communication, not a one-to-one mapping from tool events to packets.

follow from the source evidence and which need a further test.

## 2. Scope and Evidence Construction

The evidence map connects Agent execution to an observed communication signal, a hidden target and the rights needed to observe it.

### 2.1. Inclusion and retrieval

An Agent uses an LLM to select actions, with external feedback able to affect later action or termination. Client, tool, model and inter-agent interfaces are eligible. Transit observers obtain traffic metadata; contacted recipients see their addressed requests; active auditors also choose probes or challenges. Negative results qualify: PrivacyTrace's metadata/order condition remains included despite scoring below its prior reference [37]. Whole semantic trajectories or browser instrumentation alone enter the adjacent layer, while deliberate covert messages and unrelated prompt-injection defenses fall outside this question [38], [27].

The fixed eligibility window covers first public submissions from 1 January 2024 through 9 November 2026. Twelve topic queries, a native arXiv query and inclusion rules preceded the retrieval pass; title/publication checks and one-step citation follow-ups completed it. The arXiv query returned 180 records. Public-web queries supplied 33 scholarly candidates across 101 displayed URLs; deduplication yielded 209 records. Two citation leads brought screening to 211: fourteen direct, forty adjacent, 155 outside scope and two unresolved identities. The eligible CHeaT (Cloak, Honey, Trap) seed adds the fifteenth direct source (Figure 2). The SSH honeypot entered through citation tracing, illustrating the value of combining retrieval routes. Exact queries and dispositions are in Supplement S2.

Twenty-two separately selected mechanism anchors comprise sixteen published communication studies and six serving studies. They supply the established relations in the transfer matrix; the fifteen direct studies supply the Agent application evidence. These layers retain separate inclusion records and denominators. The anchor set is selected for mechanism correspondence, rather than a census of classical traffic analysis.

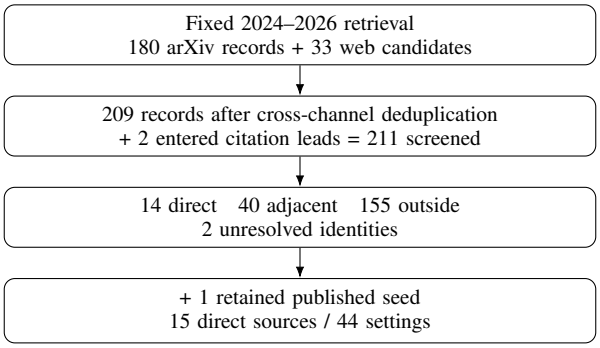



Figure 2. Executed coverage audit. Counts are deduplicated work/lead records; two identities remain unresolved. Mechanism anchors use a separate inclusion rule.

### 2.2. Extraction and source dependencies

Forty-four setting records bind source version and locator to access, features, learner, target, population, split, intervention, result and cost. Labels, execution outcomes, failure treatment and policy context are recorded alongside them. WRA and CHeaT are checked against official USENIX texts. One HTTP study has a verified ACM workshop identity, while its detailed measurements come from the extended arXiv version [52], [53]. Twelve other works lack independently verified formal publication identities at the cutoff.

Text reading, displayed-number reconstruction, selected-code inspection and data recomputation have separate entries in the extraction sheet. For example, WRA's decoy and visibility conditions are checked in the published text; OHTTP refitting is checked in selected code. Each check supports

its matched premise. Section 9.1 reports what remains when sources are removed. Extraction is author-performed with AI assistance and without independent human coding.

For the descriptive summary, every setting receives an access class and intervention purpose. Protected-condition adaptation and original-task utility are counted within the four query/workflow privacy settings. Three identity-obfuscation settings and two adversarial-Agent blocking settings have different objectives; thirty-five settings contain no protection comparison. A located item identifies evidence in the retained record. An unlocated item leaves that question open. Supplement S5 lists the assignments and source locators.

### 2.3. Position among security surveys

Broader surveys organize LLM and Agent threats by lifecycle, attack surface and defensive role [56], [57], [58]. We connect communication measurements to execution, access and protection cost. Supplement S7 gives the detailed scope comparison.

## 3. Reading the Evidence Map

The same ticket can expose different facts at the network link, the tool recipient and the model service.

### 3.1. From private task to observed execution

An execution maps a hidden task or state through an agent policy and environment into communication, then into an observer's features:

$$\text{hidden task/state} \longrightarrow \text{agent execution} \longrightarrow \text{observed communication} \longrightarrow \text{inference}.$$

Different cuts through this chain expose different objects. A client link can reveal response mode or hosted-agent identity. Tool egress exposes contacted services and request/response shape. The model link carries prompts, tool feedback and model responses. Inter-agent links expose delegation and coordination. Figure 1 places these observations on the feedback path.

### 3.2. What the inferred label means

The ticket example separates four labels. *Intent* is the requested objective or assigned task family. *Execution* is the recorded action sequence. *Success* states whether that sequence and its final state meet the task's criterion. *Authorization* states whether the action is permitted for the principal under the applicable policy. A failed refund task can still contain a refund attempt; a successful refund can still be unauthorized. An evaluation should bind its target to the corresponding reference instead of letting one label stand for all four.

Exposure is relative to both the target and the observer. The same execution may expose intended-task information on M and actual contacted services on T. Initial instructions, tool feedback and later decisions all contribute to the complete M trace. Assigning a task-family label to that trace measures intention recognition; attributing the signal to a particular feedback step requires a further controlled comparison.

### 3.3. What each record specifies

Six fields give a score its meaning: the *interface and visible representation*; *observer access*; *prior information* supplied by the evaluator; *target*; *population and base rate*; and *protected training access*. These fields define the comparison record. The matrix records the fields once; the reporting checklist in Section 8 turns them into entries for future studies.

Access determines what a feature can mean. A server can inspect decrypted HTTP/2 properties that a transit observer cannot. The content-aware MCP study's "metadata" includes argument hashes, and its graph uses content matches [45, Secs. 4.1–4.2]. Its representation therefore requires more than packet metadata. Logs and browser instrumentation can provide evaluation labels without becoming attacker inputs.

ANT's supplied stage boundary is prior information; its 2,101 stages and 47 labels define the decision population; the primitive is the target [9]. In the live OHTTP comparison, an unchanged random-forest (RF) family refits on protected data, so it is an adapted attacker [3]. These examples show how to read access, target, population, supplied boundaries and training conditions in sequence. A source setting that needs an unrepresented field would require revising the map.

### 3.4. Deployment determines who can see each signal

Table 1 maps common deployments to observation rights and the corresponding source evidence. A visible model link is an access condition, not a property granted to every attacker. Captured packets, TLS records, HTTP streams, server-sent events (SSE) and tool calls are different units. Segmentation, buffering and retransmission change their correspondence. Supplement S7 gives the protocol-level map.

Two simultaneous tickets may use separate connections or share an encrypted HTTP/2 connection. In the latter case, a transit observer sees combined packets while stream identifiers remain encrypted. A gateway that buffers SSE output changes the pattern again. The connection and flush policy therefore determine the measured trace.

The same reasoning explains defense placement. Padding a tool message before a gateway does not specify the gateway's output. Local MCP over `stdio` removes that network tool leg, while remote model requests may remain. Shared key/value (KV) cache state creates an additional timing interface for a querying tenant. The map therefore records the transport, observation layer, terminating parties

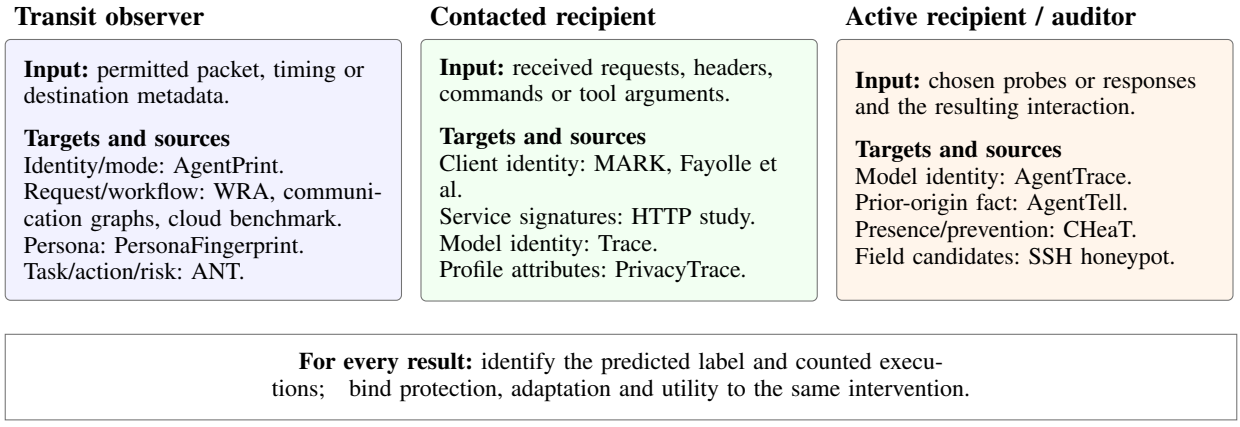



Figure 3. Evidence space by observation rights. The fifteen studies are placed by the mapped settings, not ranked by accuracy or access strength. Some offer richer additional views; Tables 3–4 identify those conditions and their metrics. Known-service signatures and field candidates are descriptive endpoints, not held-out recognition rates.

TABLE 1. DEPLOYMENT AND ACCESS MAP. THE CONFIGURATIONS ARE ANALYST-DEFINED; CITED SETTINGS TEST THE LISTED VIEW, NOT EVERY COMPONENT OF THE DEPLOYMENT. A GATEWAY'S PLAINTEXT ACCESS REQUIRES TLS TERMINATION.

| Deployment | Observer and permitted view | Existing evidence for that view | Protection consequence |
|---|---|---|---|
| Enterprise egress | A transit monitor sees routed model/tool flows, sizes, times and any exposed destination association. | WRA: supplied research-domain traces; ANT and cloud benchmark: model/combined traffic [2], [9], [4]. | Specify which services cross the monitored egress. A union of flows is available only when the monitor can observe them. |
| Shared gateway or relay | Transit sees multiplexed gateway output. A terminating gateway can additionally read addressed requests. | Live OHTTP: passive-link inference under stated relay trust [3]; PrivacyTrace: semantic recipient/gateway access [37]. | Separate relay association protection from content trust. Connection sharing changes the observation unit. |
| Local model, remote tools | An external observer sees remote tool traffic; the local model call has no corresponding external network leg. | WRA's destination-recovery mechanism is relevant to tool egress [2]; whole-deployment protection is untested in the mapped comparison. | Evaluate tool and downstream calls. Do not import a remote-model attack score into an absent network leg. |
| Cloud model, remote tools | An authorized vantage may see either or both encrypted legs. A contacted tool sees its own arguments. | ANT/cloud benchmark test model traffic [9], [4]; PrivacyTrace tests recipient-visible representations [37]. | Check permitted model and tool views after the intervention; recipients and passive monitors need different protections. |

and shared state alongside each result. Section 7.1 compares real HTTP/2 concurrency with paired feedback and shaping interventions.

## 4. Classical Conditions in Agent Execution

Established traffic-analysis results carry conditions on access, observation units, workloads and interventions. Table 2 connects those conditions to the Agent evidence. Its central distinction is between an evaluation principle that carries over and an empirical performance result that requires a matched Agent test.

### 4.1. Reading the transfer matrix

Open-world prevalence, false alarms, adaptive attacks and workload shift are established concerns [15], [49], [34], [16], [55], [19]. Localization also has direct precedents: Wang and Goldberg study traffic splitting without reliable page boundaries; BAPM and ARES address mixed multi-tab traces [48], [46], [47]. WF is neither inherently deterministic nor restricted to one link. Here these issues interact with feedback across independently controlled services and cover executed as application work.

*Retained* means that an established evaluation obligation still applies; *extended* identifies an additional Agent execution condition; *access mismatch* prevents importing a result under the proposed observer; and *untested* means that the mapped evidence leaves the specific performance claim open. These relationships are separate from the number and publication status of supporting sources. Distributed tracing illustrates access mismatch: WAP5 and vPath use host instrumentation [32], [33], which can label an experiment without becoming a passive attacker's input.

TABLE 2. TRANSFER OF ESTABLISHED CONDITIONS TO AGENT EXECUTION. RELATIONSHIP LABELS CONCERN THE STATED MAPPING; EVIDENCE STRENGTH AND MISSING TESTS APPEAR SEPARATELY. A MISSING AGENT TEST DOES NOT REFUTE THE CLASSICAL RESULT.

| Established relation and condition | Agent mapping / consequence | Relation | Agent evidence and remaining test |
|---|---|---|---|
| Unknown boundaries and background affect event recognition [48], [46], [47], [35]. | Tool bursts, model waits and task intervals are different units. Evaluate proposed events with false alarms as well as classification. | Retained | ANT supplies stage boundaries; the local concurrent study tests particular detectors. Greater Agent localization difficulty is untested. |
| Defense evaluation must allow adaptation to the protected distribution [34], [16], [19]. | Record protected training access and still-visible links; an unchanged learner family can refit. | Retained | Transformed and live OHTTP settings report adaptation in one preprint [3]; WRA's matched protected examples remain unresolved. |
| Coarse timing/volume can remain useful under specified padding [34]; serving streams expose length/gap signals [20], [12]. | Name the shaping policy, receiving hop and Agent workload before carrying over a measured effect. | Untested generaliza-tion | Cloud and ANT measure model-view recognition [4], [9]. A general protected-Agent SSE claim is not tested by these observations. |
| An observation result is tied to the visible paths and representations [30], [36]. | Feedback forwards tool information into later model requests across separately controlled services; evaluate every permitted residual view. | Extended | Published WRA tool-side evidence and preprint model/recipient results establish different exposed interfaces. Causal feedback attribution needs its own test. |
| STP tunnel endpoints remove cover before it reaches devices or servers [36, Sec. 7.1]. | When cover executes as research, add original-task resource, state and feedback constraints to the traffic budget. | Extended | Published WRA reports five executed research decoys; its task-utility scores belong to another intervention [2]. Harmful writes are not a WRA result. |
| Host traces can reconstruct distributed request paths [32], [33]. | Instrumentation can provide ground truth; a transit attacker must infer from its own permitted observations. | Access mismatch | ANT stage labels and local task intervals are evaluator-provided. Host tracing alone cannot establish passive encrypted-stream localization. |

Five questions locate the resulting evidence: signal mechanism, supplied-unit recognition, named deployment transfer, stream localization and protection. They are distinct tests. A reported score binds a target to an observation and a tested deployment; Tables 3–4 preserve those bindings across the field.

### 4.2. Task boundaries and execution changes

AgentPrint supplies browser interactions, WRA research traces, the communication-graph study workflows, the cloud benchmark execution timestamps, and ANT runtime-defined stages [1], [2], [3], [4], [9]. ANT's 47-group primitive test scores 0.355–0.462 accuracy on 2,101 supplied segments. Its episode split separates parent executions, while the supplied boundaries define a recognition test. Event localization additionally requires recall and false events on proposed intervals.

PersonaFingerprint keeps one session's windows in one partition [5]. A neighboring published study classifies non-overlapping 5,000-packet windows without page markers, primarily testing human browsing with Agent-generated training traces [43]. PINGPONG and voice-invocation work demonstrate detection against background traffic [35], [22].

Transfer names what changes while the target remains fixed: wording, session, composition, topology, scaffold or model. Reported composition, topology and scaffold losses concern different axes (Section 5.3). Simultaneous Agent tasks and model waits also change how tool bursts relate to complete executions. Our live comparison separates supplied-window inference from localization and retains physical connection identities.

## 5. Two Execution Dependencies for Protection

The field map connects two properties of Agent execution to defense design: information passes between services, and cover can become application work. Table 5 states the observation conditions under which particular signals matter.

### 5.1. Feedback joins separately observable services

The support-ticket execution exposes a selected tool destination, arguments received by that tool, a later model request and the model's response. WRA recovers query meaning from contacted domains under its name-association and supplied-trace conditions [2]. ANT, a network-traffic benchmark for Agent behavior, and the cloud benchmark measure task recognition from model traffic [9], [4]. The cloud benchmark reports ten-class balanced accuracy .853/.860 for graph/star coordination with all 247 features, and .540/.587 when size features are excluded and 100 of the 134 remaining features are selected. Both feature availability and selection change in that comparison. ANT's model-view values reconstructed from rounded figures remain approximate quantities in Supplement S2.

TABLE 3. ATTACK FIELD MAP, PANEL A: TRANSIT OBSERVATIONS. SCORES RETAIN THEIR METRIC, TASK AND SUPPLIED UNIT; THEY ARE REPRESENTATIVE REPORTED OPERATING POINTS, NOT A CROSS-PAPER RANKING. THE MATRIX RECORDS ALL FORTY-FOUR SETTINGS. P: INSPECTED PREPRINT; V: INSPECTED PUBLISHER VERSION.

| Interface / observer | Work / target | Observable features / learner | Evaluation unit | Reported result and boundary |
|---|---|---|---|---|
| Client–agent; transit | AgentPrint [P] [1]; hosted agent | Directional packet counts/bytes in 1,800 relative-time bins; CNN | Supplied browser interaction | Mixed-flow macro-F1: .8660 closed, .8477 open; 50 monitored agents. Behavior-F1 conflict is kept separate. |
| Agent–Web; transit | WRA [V] [2]; research query | Contacted domains and timing; ICL or fine-tuned LLM | Supplied research trace | Functional query recovery .770 with domains; .725 with domains plus timing in the compared visibility views. This is a reconstruction score. |
| Inter-agent / tools; transit | Communication graphs [P] [3]; workflow | Message/graph, size and timing descriptors; random forest | Workflow record | Nine-class accuracy .67; within-label composition holdout .18. The live defense test instead has six classes. |
| Agent–model; transit | Cloud benchmark [P] [4]; assigned task | 247 traffic descriptors; random forest | Logged successful run | Ten-class balanced accuracy .853/.860 for graph/star coordination. Size-excluded feature protocols score .540/.587. |
| Agent–Web; transit | PersonaFingerprint [P] [5]; browsing persona | Signed lengths/interarrival times; CNN representation and prediction heads | Fixed packet window; session split | Per-site macro-F1 about .78–.91; pooled-summary discrepancy excluded from synthesis. Explicit other-class tests have a different population. |
| Combined egress, M or remaining links; transit | ANT [P] [9]; task, primitive, risk | Direction/length/time; 13 learners including DF and TikTok | Supplied episode or logged stage | Five-task accuracy .431–.727; 47-primitive accuracy .355–.462. Risk test: 20 positive/189 negative, requiring recall and false alarms. |

Recipient access changes the inference problem. PrivacyTrace's tool metadata/order yields attribute-inference ASR 14.53%, below its 17.97% schema-prior reference; adding arguments yields 27.71% [37]. MARK's combined F1 .971 also uses browser behavior, while its protocol/header features are available at the contacted website [6]. Encryption against transit observation does not hide an addressed request from its recipient.

These results identify where task information is observable. They lead to a testable question: with the initial instruction fixed and task-bearing feedback released later, how much recognition remains on the model link under a stated tool-side policy? Section 7 tests this question with constructed schemas and reports both the first-feedback and complete-execution views. Initial instructions, retained task fields and model responses are distinct possible contributors to the latter.

Defense consequence. Enumerate the services and representations available to the attacker, then evaluate the views remaining after an intervention. A joint observer can use a single-link decision rule, even when a particular fitted joint classifier scores lower. At the application boundary, minimize forwarded context while preserving the information needed for the task; at the transport boundary, evaluate visible sizes, timing and association. Their effects belong to separate contrasts.

## 5.2. Cover termination determines the execution constraint

Transport cover can terminate before business execution. Executed cover performs work and may affect resources, state or later decisions. The difference determines which utility must accompany a privacy measurement.

WRA's five research decoys reduce functional query recovery from .770 to .565 [2, Sec. 8.1]. Its utility scores 8.55 and 8.54 instead compare 100% and 10% destination visibility. Thus the utility observation does not describe the decoy intervention that produced the recovery reduction. WRA establishes additional research execution; it does not report harmful business-state writes. Matched protected-condition attacker examples also remain unresolved in the selected evidence.

The live OHTTP study refits a random forest on each protected capture and pairs six-class accuracy .30 with approximately thirteen-fold bytes and eleven-fold latency [3, Sec. 9.7]. Its separate nine-class .67 score belongs to another comparison. This design point binds adapted inference to transport cost; matched original-task utility was not located.

Defense consequence. An executed-cover design needs a semantic constraint alongside a traffic budget. It can terminate decoys before business execution, or isolate their resources, state and planner feedback. These are testable design choices: measure adapted inference, additional work and original-task completion on the same protected runs. The published WRA text supports the intervention/utility distinction; the OHTTP refit/cost example depends on one preprint.

TABLE 4. ATTACK FIELD MAP, PANEL B: CONTACTED RECIPIENTS AND ACTIVE AUDITORS. THESE OBSERVERS SEE REQUESTS ADDRESSED TO THEM OR ISSUE PROBES; THEIR RESULTS DO NOT MEASURE ENCRYPTED-TRANSIT INFERENCE. E: INSPECTED EXTENDED PREPRINT WITH A SEPARATELY VERIFIED PUBLISHED SHORT VERSION. NR: NO RECOGNITION RATE REPORTED IN THE MAPPED SETTING. ASR: ATTACK SUCCESS RATE.

| Interface / access | Work / target | Observable features / learner | Reported result and evaluation boundary |
|---|---|---|---|
| Web recipient | MARK [P] [6]; client identity | TLS/HTTP2 and headers; decision tree | F1 .725 for TLS/HTTP2, .799 for headers; .971 combined with browser instrumentation, a richer view. First-pass trials. |
| Web recipient | Fayolle et al. [P] [7]; client / access | Protocol signatures; richer behavior classifier and site challenges | Fingerprint analysis excludes failed visits. Access success is separately verified through form/cookie/page identifiers; not a privacy score. |
| Documentation recipient | HTTP signatures [E] [52]; service/client | Request method/version, headers and origin; descriptive comparison | Nine coding agents and six assistants, three trials each; NR. Request signatures are observations, not a held-out identification rate. |
| Terminal recipient | Trace [P] [51]; model family | Command TF-IDF unigrams/bigrams; linear SVM | Seven families; 2,028 clean/injected sessions: macro-F1 .981; held-out scaffold .815. A separate proprietary-scaffold test reports 78% accuracy. |
| Tool interface; active audit | AgentTrace [P] [50]; model identity | Serialized arguments, parser errors and repair order; Markov likelihood | Six models, 50 held-out probes per trial: 96.0% accuracy; 10 probes: 92.8%. Repeated stochastic trials reuse probe identities. |
| Web recipient; active choice | AgentTell [P] [8]; prior-origin fact | Which offered action is requested; loaded/cold comparison | Matching choice in 61.1% of 7,630 loaded sessions; the leakage measure subtracts cold matching. Actions share a completable probe task. |
| Tool recipient / gateway | PrivacyTrace [P] [37]; profile attribute | Tool metadata/order; content-added LLM inference | Metadata ASR 14.53%, schema prior 17.97%; arguments 27.71%. 68,000 attribute slots are clustered within 4,000 trajectories. |
| Contacted asset; active challenge | CHeaT [V] [10]; agent presence | Planted URL following; separate password-output control | Presence-detection mechanisms, NR for natural-traffic classification. Capture-the-flag prevention is a distinct endpoint (Supplement S7). |
| SSH recipient; active probes | LLM Agent Honeypot [P] [54]; possible agent | Prompt compliance and response-time heuristic | Three-month deployment: eight candidate agents among 8,130,731 interactions. No externally verified field labels or detection precision/recall. |

### 5.3. Execution changes delimit the two findings

A private recurring routine can be recognizable while its classifier transfers poorly to a new execution organization. In the communication-graph study, composition holdout keeps the nine workflow labels but changes their capability combinations: accuracy changes from about .67 to .18, with zero recall for seven classes [3, Sec. 9.12]. The cloud benchmark's graph/star change separately reports about .83 to .52 under its corresponding feature protocol [4]. Each axis depends on one preprint.

Implementation and failure handling can also change the observation. Trace reports macro-F1 .981 in its mixed-session model-attribution test and .815 on scaffold holdout, retaining 1,010 of 1,050 planned clean runs [51]. AgentTrace's active probes reach 66.8% attribution from initial outcomes and 96.0% from ordered repairs at fifty probes [50]. Excluding failed sessions and observing recoverable errors are different operations. These source-specific conditions delimit what an attack or defense says about new workloads; they do not estimate a common filtering effect.

Finally, a traffic target retains its own security meaning. ANT labels benchmark risk, AgentTell tests action choice under a disclosure rule, and CHeaT measures prevention of a defined CTF objective [9], [8], [10]. Authorization additionally requires a principal and policy context. Two refunds that differ only in an unobserved approval can have identical traffic. This conditional argument identifies the extra policy input an authorization monitor needs.

## 6. Defense Mechanisms and Execution Costs

Defense placement determines both the signal changed and the party that must enforce the change.

### 6.1. What a defense changes

Table 6 organizes six intervention choices by enforcement point, trust, cost and residual signal. Association defenses hide a destination or its link to a user. Padding changes sizes, while scheduling changes temporal structure. Transport cover can terminate before business execution; execution cover adds real work. State isolation limits shared-resource signals, and content minimization limits what a recipient learns. These choices can be combined, but their costs and remaining views must be evaluated together.

TABLE 5. CONDITIONAL RISK MAP. A ROW APPLIES WHEN ITS ACCESS AND EXECUTION CONDITIONS HOLD. THE FINAL COLUMN STATES A PROTECTION PRIORITY OR DESIGN HYPOTHESIS, NOT A DEMONSTRATED MITIGATION. SCORES, TARGETS AND RIGHTS DIFFER ACROSS ROWS, SO THERE IS NO GLOBAL DANGER RANKING.

| Interface and signal | Condition and evidenced inference | Protection consequence |
|---|---|---|
| **Tool egress:** selected domains and timing | The observer can associate contacted names with a supplied research trace. WRA recovers query meaning; linked histories use a separate persona endpoint [2]. | Prioritize destination association when the private target is research intent. Reduced name visibility needs a residual traffic assessment; executed decoys need task-utility constraints. |
| **Model link:** lengths, directions and timing | Model traffic is visible and associated with the evaluated task. ANT and the cloud benchmark measure task information in this view [9], [4]. Initial instructions and later feedback coexist. | Tool-only protection leaves this view to be assessed. Compare model-only and joint adaptive attacks; isolate feedback before choosing a feedback-specific defense. |
| **Coordination:** message graph, volume and sequence | Workflow records or coordination traffic expose execution organization. Composition and topology transfer tests change that organization under known task labels [3], [4]. | Target observable scheduling and association, not just one payload size. Evaluate changed compositions and topology separately from recurrent-workflow privacy. |
| **Client link:** packet counts, directions and bursts | AgentPrint's supplied browser interactions support hosted-agent identity inference [1]. The result presumes those observation units and identities. | Client-facing traffic needs its own protection assessment even if tool links are hidden. Task discovery and new identities require separate tests. |
| **Contacted recipient:** headers, commands or arguments | MARK, Trace and PrivacyTrace use recipient-visible representations. In PrivacyTrace, metadata alone is below its prior, while adding arguments improves attribute inference [6], [51], [37]. | Minimize sensitive request content and assess implementation signatures. Transport encryption does not conceal the recipient's intended plaintext. |
| **Active probing:** challenges, choices and repairs | A party can elicit responses and observe later behavior. AgentTrace attributes models from repairs; AgentTell observes a choice under a disclosure rule [50], [8]. | Test error normalization and restrictions on cross-context disclosures while retaining task recovery. These hypotheses need active-access evaluations. |

### 6.2. Relating communication defenses to Agent execution

STP illustrates transport cover that can terminate at a tunnel endpoint [36]; Walkie-Talkie and FRONT provide shaping mechanisms [17], [18]. For the ticket Agent, this could mean padding the tool response at an egress proxy, pacing model tokens, or sending cover to a trusted terminator. Calling the refund tool as a decoy would instead cross the business-state boundary. The intervention must therefore specify both where cover is removed and what work it can trigger.

The measured design points occupy different parts of this space. WRA changes executed research, live OHTTP changes destination association, and the paired forwarding study changes the context sent to the model. Section 5 gives the source-specific recovery and cost results; Section 7 tests forwarding and connection policies. Supplement S7 retains the complete defense map, including identity obfuscation and adversarial-Agent blocking as distinct objectives.

### 6.3. Choosing a layer in practice

Start from the private object and the observer's rights. Protect destination associations when contacted domains expose the query; protect visible model output sizes and timing when those carry the target; minimize sensitive arguments when the recipient is the observer. Then test what remains on other permitted links. A shared gateway changes which tasks contribute to a connection, so its connection policy belongs in the evaluation. For executed cover, bind privacy, token/tool use and original-task state to the same protected runs. These choices turn the risk map into a deployment-specific protection plan.

## 7. Paired Feedback and Concurrent Execution

The mechanism study compares forwarding at the first feedback step with exposure over the complete execution. Concurrent conditions test task discovery under actual connection sharing. Its constructed schemas specify a dependency to examine; the source synthesis supplies the executed-cover finding.

### 7.1. Workload, framework and paired interventions

A LangGraph state graph [59] alternates model decisions and actual MCP tool calls. Every execution begins with

TABLE 6. DEFENSE DESIGN SPACE: ENFORCEMENT AUTHORITY, COSTS TO MEASURE AND REMAINING EXPOSURE. SOURCE-SPECIFIC MEASUREMENTS APPEAR IN THE TEXT AND SUPPLEMENT S7.

| Layer / mechanism | What must be trusted or controlled | Costs to measure together | Residual exposure and practical decision |
|---|---|---|---|
| Naming / association: encrypted name resolution, relays, OHTTP | Name resolver or relay placement; the parties able to link the user to the destination | Added hops, bytes, latency and availability | Shape can survive name hiding. Choose this layer when destination association is the private target; then test remaining observers. |
| Representation: message, TLS-record or packet padding | Component that controls the final visible size; padding survives later encoding | Added bytes, record/packet count, fragmentation | Timing, count and other links remain. State the padding layer and test downstream captures. |
| Scheduling: batching, pacing, fixed-rate cover | Sender/relay controls output schedule and cover lifetime | First-response delay, completion delay, bandwidth during idle periods | Admission, termination and overflow can reveal activity. Budget these events as part of the visible trace. |
| Transport cover: dummy exchanges discarded at a tunnel endpoint | Trusted cover terminator; no dispatch into business tools | Cover bandwidth, queueing, resource use | Other endpoints still see their inputs. Prefer this boundary when the aim is traffic shaping without extra business work. |
| Execution cover: extra research or tool calls | Tool privileges, isolated state and whether decoy feedback reaches the planner | Model tokens, tool quota, task success, state changes and latency | A decoy can affect the real execution. Measure privacy and original-task utility on the same protected runs. |
| Shared-state isolation / minimization | Tenant separation, cache keys and information passed between services | Lost cache reuse, compute, latency, task accuracy | Isolation removes a sharing channel, not every traffic signal. Minimize tool arguments and feedback according to each recipient's role. |

the same support-queue instruction. The first tool response supplies one of six requested service actions: refund, address update, escalation, replacement shipment, cancellation or receipt. Tool responses contain service-specific supplementary records with randomized contents and counts. These schemas deliberately expose a testable category dependency; they do not sample its prevalence in deployed tools. Tasks require one to three independently assigned checks; a fixed random subset encounters one recoverable check failure. The framework exposes unfinished prerequisites before permitting all six business actions, leaving the action choice to the model. Applied fixture state and goal completion are recorded separately.

The plan assigns 312 task instances to 64 sessions per condition, with 0–10 arrivals per session. Labels are balanced over train/validation/test partitions while session compositions vary. The 32/8/24 session split contains 156/36/120 tasks. Test requests use four held-out phrasings per action. Empty sessions, missing classes and duplicate same-class tasks prevent class occupancy from being inferred from a fixed block composition. Live MCP background requests continue during idle periods.

Five conditions reuse each task's fixture and seed. *Full* forwards the entire structured tool result to later model requests. *Projected* forwards a fixed allowlist of task and prerequisite fields, dropping supplementary records after the unchanged response reaches the client. Both run serially with padded tool messages. *Dedicated* and *Shared* run four workers with separate or shared connections per service. *Joint* additionally pads model requests and shapes model SSE frames under the shared policy. Incremental frame delivery can exceed its minimum schedule; this is measured, not treated as constant-time protection.

Qwen3:8B and Llama3.1:8B use fixed local weights. MCP Streamable HTTP and the model gateway use TLS 1.3/HTTP/2; the model gateway streams SSE. Packet capture observes encrypted sizes, timing, direction and physical connection grouping. Private tool logs and decrypted HTTP stream identifiers supply verification and ground truth only. The supplied-window analysis aggregates every visible connection within its window; continuous detection receives session boundaries and connection identities without task or private stream assignments.

The local cohort completed 3,120 executions and 13,680 model requests. Wire verification confirms overlapping model streams in 107 of 128 sessions in each shared condition, with up to four simultaneous requests on one connection. The remaining sessions include empty and single-task cases.

## 7.2. Inference, localization and uncertainty

Primary attacks are standardized logistic regression (LR) and the released 175-feature k-FP extractor [49] with a 1,000-tree random forest. An aggregate random forest and temporal convolutional classifier provide additional attack families. Validation chooses hyperparameters and checkpoints before test scoring. Prefix, first post-ticket and full-execution views separate pre-feedback recognition from later inference. Paired Full–Projected comparisons measure the forwarding intervention; Dedicated–Shared comparisons

address connection policy. Phase attribution is specified for the serial pair. Shared and Joint use complete windows and continuous detection: a task's post-ticket interval can contain other tasks' model streams, so a score for that interval would mix their exposure.

Continuous multilabel LR and random-forest detectors use validation-selected windows and thresholds. A gap-based detector and constant class-presence controls remain explicit baselines. Event scoring matches distinct task instances, including repeated classes, at overlap thresholds .3/.5/.7. Mean false events per session accompany recall; the supplement retains duration-normalized rates. Uncertainty resamples complete sessions. LR additionally refits under training/test session resampling, conditional on the selected family and designed workload. Eight primary local contrasts form one Holm family; exploratory and remote comparisons remain separately identified.

### 7.3. First feedback and complete-execution exposure

Table 7 places the phase-level intervention beside the complete-execution endpoints. The six-class majority reference is 20/120 locally. Under Full forwarding, first post-ticket LR scores are 43/120 for Qwen and 40/120 for Llama, versus 25/120 and 19/120 before feedback. Projected forwarding gives 23/120 and 22/120. The Full–Projected point differences are 16.7 and 15.0 percentage points; unadjusted training/test-session refit intervals are $[4.0, 29.8]$ and $[3.1, 32.8]$ points.

All eight primary local comparisons exceed .05 after exact Holm adjustment, including every post-ticket–prefix contrast. Full–Projected LR values are .0778 and .1172; the smallest adjusted value, .0508, compares Llama's Full post-ticket and prefix phases. The unadjusted refit intervals answer a different, conditional uncertainty question. Exploratory RF scores are 50/120 versus 30/120 for Qwen and 59/120 versus 19/120 for Llama. CNN scores are 20/120 versus 20/120 and 26/120 versus 20/120. Table 7 reports all four families. Their achieved scores provide attack evidence, not a leakage upper bound; the strongest test score does not replace the prespecified primary attacks.

Complete-execution outcomes depend on the model and attack. Model-only LR changes from 46 to 44 correct for Qwen and from 62 to 33 for Llama; RF changes from 61 to 31 and from 73 to 34. Forwarding minimization leaves task-relevant fields and model responses in the execution. Their separate contributions remain unmeasured. Each local condition has 518 held-out tool calls, with equal call counts in every task pair; call order or action differs in 35 Qwen and 26 Llama pairs. Thus the contrast bundles content, encoded length and execution order, while the observed Llama drop is not explained by fewer tool calls. Supplement S1 gives the path audit and saved prefix diagnostics, including Qwen LR's unadjusted permutation $p = .0595$ for 25/120.

### 7.4. Concurrency, event detection and cost

Concurrency changes task association as well as traffic. Complete-window attacks are weaker under both Dedicated and Shared conditions, whose window representations pool all visible connections during overlapping executions. Their serial-to-concurrent decline consequently combines connection policy with window mixing.

Table 8 reports the prespecified IoU .5 endpoint beside .3/.7 sensitivity checks, using the same saved event predictions. Under Shared, LR/RF recover 35.0–43.3% of instances at .5 with 4.3–18.6% precision and 8.21–44.79 false events per test session, including empty sessions. At .3, all-active exceeds both learned Shared detectors in F1 for both models: .295 versus .216/.106 for Qwen, and .508 versus .333/.464 for Llama. At .5, the learned detectors exceed that constant baseline. This combination of coarse predictions and a permissive criterion rewards session occupancy. At .5, tighter intervals matter, yet false alarms remain high. These are diagnostics of the selected detectors and overlap rule; they provide neither a deployment prior nor precision at a common recall target.

Held-out goal completion is 119/120 for Qwen in every condition, and 117/120 for Llama except Projected at 118/120. Projection's model-byte ratio is about .78 of Full in both local cohorts. Joint shaping costs 19.5 and 22.5 times Shared's model-link bytes, while model-only LR changes from 25 to 18 and from 19 to 25 correct. Of 2,736 shaped model responses, 1,818 exceed the forty-frame minimum. Joint also improves Qwen's event RF score despite shaping (Table 8); the present contrasts do not isolate why. These mixed outcomes and variable lengths make this an observed design point, not a privacy–cost frontier. Supplement S1 gives complete completion, timing, bytes and uncertainty results.

### 7.5. Separate hosted-model retest

The hosted cohort uses 96 tasks per condition, with 36/12/48 training/validation/test decisions and sixteen nonempty test sessions. Its mutable alias and tunnel observation change the backend and network together. All 192 attempts have terminal outcomes; five invalid model outputs remain in the recognition denominators.

Against the 8/48 majority reference, LR scores 5/48 before feedback, 15/48 after Full feedback and 10/48 after Projected feedback. k-FP scores 3/48, 6/48 and 7/48; its Full–Projected direction reverses. None of the four primary Holm-adjusted comparisons crosses .05. The LR difference is 10.4 points with an unadjusted training/test-refit interval of $[-3.2, 41.5]$. Complete-execution model-only LR is 9/48 in both conditions (Table 7). The retest supplies no consistent intervention effect across the declared attackers and endpoints.

Held-out completion is 46/48 for Full and 47/48 for Projected. The model-byte ratio is .932 with paired interval $[.868, 1.003]$. Supplement S1 reports this cohort separately from the local study.

TABLE 7. CONDITION-MATCHED RECOGNITION: CORRECT PREDICTIONS OUT OF $n$ (SIX-CLASS REFERENCE $n/6$). PREFIX IS THE INITIAL FULL MODEL CALL; POST IS ITS FIRST CALL AFTER THE TICKET. COMPLETE SPANS ALL REQUESTS AND RESPONSES. RF AND CNN ARE EXPLORATORY (E); NEITHER REPLACES LR/K-FP IN THE PRIMARY FAMILY. $p_{proj}$ IS THE EXACT HOLM-ADJUSTED FULL–PROJECTED POST TEST. COMPLETE-VIEW COMPARISONS ARE DESCRIPTIVE. SUPPLEMENT S1 REPORTS ALL UNCERTAINTY AND FROZEN-TRANSFER RESULTS.

| Executor | Attack | $n$ | Prefix | Post M | | Complete M | | Complete T+M | | $p_{proj}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Full | Proj. | Full | Proj. | Full | Proj. | |
| Qwen3:8B | LR | 120 | 25 | 43 | 23 | 46 | 44 | 54 | 39 | 0.0778 |
| Qwen3:8B | kFP | 120 | 18 | 32 | 20 | 43 | 39 | 39 | 30 | 0.1989 |
| Qwen3:8B | RF (E) | 120 | 25 | 50 | 30 | 61 | 31 | 55 | 30 | — |
| Qwen3:8B | CNN (E) | 120 | 20 | 20 | 20 | 22 | 22 | 20 | 22 | — |
| Llama3.1:8B | LR | 120 | 19 | 40 | 22 | 62 | 33 | 53 | 29 | 0.1172 |
| Llama3.1:8B | kFP | 120 | 14 | 31 | 20 | 47 | 34 | 45 | 31 | 0.1989 |
| Llama3.1:8B | RF (E) | 120 | 24 | 59 | 19 | 73 | 34 | 55 | 30 | — |
| Llama3.1:8B | CNN (E) | 120 | 20 | 26 | 20 | 22 | 15 | 19 | 18 | — |
| Hosted alias | LR | 48 | 5 | 15 | 10 | 9 | 9 | 12 | 6 | 0.9214 |
| Hosted alias | kFP | 48 | 3 | 6 | 7 | 9 | 5 | 13 | 11 | 1.0000 |
| Hosted alias | RF (E) | 48 | 8 | 15 | 9 | 7 | 7 | 13 | 8 | — |
| Hosted alias | CNN (E) | 48 | 8 | 11 | 5 | 10 | 8 | 9 | 11 | — |

TABLE 8. EVENT OVERLAP SENSITIVITY ON THE SAME PREDICTIONS. EACH CONDITION CONTAINS 120 TASKS IN 24 SESSIONS, INCLUDING FOUR EMPTY SESSIONS. LEARNED DETECTORS USE T+M; ALL-ACTIVE EMITS ONE SESSION INTERVAL PER CLASS. IOU .5 IS PRIMARY; .3/.7 ARE SENSITIVITY CHECKS. FP/SESSION DIVIDES ALL FALSE EVENTS BY 24, RETAINING EMPTY SESSIONS. SUPPLEMENT S1 RETAINS HOURLY NORMALIZATION, COUNTS AND INTERVALS FOR THESE BOUNDED RECORDINGS.

| Executor | Policy | Detector | F1 (.3) | F1 (.5) | F1 (.7) | P (.5) | R (.5) | FP/session (.5) |
|---|---|---|---|---|---|---|---|---|
| Qwen3:8B | Shared | LR | 0.216 | 0.119 | 0.048 | 0.072 | 0.350 | 22.54 |
| Qwen3:8B | Shared | RF | 0.106 | 0.077 | 0.035 | 0.043 | 0.400 | 44.79 |
| Qwen3:8B | Shared | all-active | 0.295 | 0.038 | 0.000 | 0.035 | 0.042 | 5.79 |
| Qwen3:8B | Joint | LR | 0.196 | 0.117 | 0.050 | 0.070 | 0.375 | 25.04 |
| Qwen3:8B | Joint | RF | 0.329 | 0.188 | 0.090 | 0.123 | 0.400 | 14.29 |
| Qwen3:8B | Joint | all-active | 0.265 | 0.038 | 0.000 | 0.035 | 0.042 | 5.79 |
| Llama3.1:8B | Shared | LR | 0.333 | 0.216 | 0.104 | 0.144 | 0.433 | 12.88 |
| Llama3.1:8B | Shared | RF | 0.464 | 0.249 | 0.105 | 0.186 | 0.375 | 8.21 |
| Llama3.1:8B | Shared | all-active | 0.508 | 0.098 | 0.000 | 0.090 | 0.108 | 5.46 |
| Llama3.1:8B | Joint | LR | 0.441 | 0.251 | 0.104 | 0.186 | 0.383 | 8.38 |
| Llama3.1:8B | Joint | RF | 0.429 | 0.274 | 0.129 | 0.199 | 0.442 | 8.92 |
| Llama3.1:8B | Joint | all-active | 0.530 | 0.121 | 0.000 | 0.111 | 0.133 | 5.33 |

## 8. Evidence Availability and Reporting

Attack adaptation and original-task utility occur in different mapped protection settings. Table 9 identifies four eligible query/workflow privacy comparisons: two document adaptation and one documents matched original-task utility; none locates both. The adapted settings share one preprint, while the utility observation is WRA's published visibility comparison.

Different objectives need different denominators. Three identity-obfuscation settings, two adversarial-Agent blocking settings and thirty-five without a protection comparison remain outside this denominator. The access map separately counts twenty-two transit settings, ten recipient settings, eleven active settings and one downstream preference analysis. These author-assigned classes partition settings; source groups overlap. The counts describe the retained corpus.

A compact reporting contract. Arp et al. and WF guidelines establish sampling, labels, observation, splits and adaptive evaluation [55], [15], [16]. Our eight-field checklist adds service/access mapping, label stage and original-task costs. Supplement S5 provides the checkable form, guideline correspondence, eligibility ledger and completed WRA, OHTTP and ANT examples. Each entry records evidence, a missing item or inapplicability.

The reporting unit is the intervention: which observer remains, how the attack adapts, and what happens to the original task. Recording these together would close the mapped privacy–utility gap.

## 9. Synthesis and Research Agenda

### 9.1. Source dependencies of the two findings

The transfer matrix separates established conditions from evidence for their Agent application. Model- and tool-side task observations draw on several studies; the decoy/utility mismatch comes from WRA's published text. Live OHTTP refitting and each composition, topology or scaffold transfer axis depend on individual preprints. The

TABLE 9. EVIDENCE LOCATED IN THE RETAINED RECORDS. COUNTS USE ELIGIBLE SETTINGS; THE SECOND COLUMN ALSO GIVES DISTINCT SOURCES. "NOT LOCATED" LEAVES EVIDENCE UNRESOLVED WITHIN THE INSPECTED SCOPE. THESE COUNTS DESCRIBE AVAILABILITY, NOT STUDY QUALITY OR CHECKLIST VALIDITY.

| Question | Eligible settings / sources | Located | Supporting settings and consequence |
|---|---|---|---|
| Protected-condition attack adaptation | 4 / 2 | 2/4 | Communication graphs: transformed captures and live OHTTP. Both come from one preprint. |
| Original-task utility under the same intervention | 4 / 2 | 1/4 | WRA visibility comparison. It does not price WRA's executed-decoy condition. |
| Adaptation and original-task utility in one setting | 4 / 2 | 0/4 | Adaptation and utility are located in different records; transport cost alone does not fill the task-outcome field. |
| Original-task utility for executed cover | 1 / 1 | 0/1 | WRA's five research decoys report recovery and extra research; matched original-task utility is not located. |

two located adapted privacy settings share one source. These dependencies matter more than treating every setting as an independent confirmation.

Source withdrawal tracks thirty-three statements across seven comparisons. Supplement S2 gives the complete summary and twenty removal scenarios, including individual sources, shared-author groups and publication filtering.

The strict published-text subset retains WRA and CHeaT: query recovery, the decoy/visibility distinction and a defined prevention outcome survive. Other label, population and policy contrasts depend on the wider corpus. Removing one transfer source removes evidence for that axis, rather than changing the other axes. The empirical feedback intervention supplies a separately identified controlled case; it does not turn source-dependent findings into replications.

### 9.2. Tests that would change the protection decision

The next studies should measure the information and execution that cross a defense boundary. Table 10 gives four concrete comparisons; Supplement S6 supplies their source basis. The feedback intervention tests one controlled path. Longer histories, different tools, stronger forwarding minimization and production gateways can alter its outcome.

For practitioners, three decisions follow. First, map a private target to the model, tool and recipient observations actually available to the adversary. Second, check adapted joint and permitted single-link attacks after shaping or forwarding changes. Third, if cover executes, specify which task resources, state and feedback it may affect, then measure those quantities on the protected runs. A design that lowers inference while meeting these application constraints would add a useful point to the sparse defense map.

## 10. Validity, Ethics and Availability

Evidence scope. Source locators and withdrawal analysis expose dependencies; index failures and query misses limit coverage. Extraction is author-performed with AI assistance, without independent human coding. The source-specific measurements and our controlled tasks represent different populations.

The paired mechanism study covers six synthetic actions with deliberately category-bearing supplementary records on one host. All adjusted primary tests exceed .05. The forwarding intervention changes content, encoded length and execution order together; equal tool-call counts do not isolate the first two. Complete-execution and overlap-threshold contrasts are reported alongside the focal phase results. Session resampling, including LR refitting, measures variation within this workload and selected learner family. Recognition receives phase or task windows; localization receives session boundaries and synthetic live background traffic. These conditions leave generalization to other workloads, enterprise background and gateways open. The remote retest changes the backend and network together and uses an alias without pinned weights.

Earlier serial, common-prefix and fixed-composition studies are archived as separate populations in the review artifact, with their failures, inconclusive results and unknown responses. Their outcomes are not pooled with the present experiment.

Ethics and assistance. Inputs are synthetic, business actions update isolated fixtures, and capture filters restrict collection to the experiment. Remote model use has the account holder's authorization; provider-specific research permission and applicable terms remain unverified. Known failures and unknown responses are retained separately. No human-participant experiment, customer operation or hosted-provider vulnerability finding enters the study. AI tools assisted source extraction, implementation and writing; saved observations and executable analyses support the reported measurements.

Traffic inference can enable unwanted profiling as well as monitoring. The review package therefore contains synthetic, payload-free metadata, features, predictions and offline analyses. Credentials, private transcripts, raw captures and third-party full texts are excluded. Reusable analysis methods still carry misuse potential. A protection summary should report adapted attacks, remaining views and original-task outcomes together; a frozen attack's decline alone can overstate the measured protection.

TABLE 10. TESTABLE OPEN QUESTIONS DERIVED FROM THE SYNTHESIS. Q1 AND Q3 SPECIFY PROSPECTIVE TESTS. Q2 EXTENDS PAIRED FORWARDING TO SEPARATE ITS MEDIATORS; Q4 EXTENDS LIVE DETECTION TO NEW WORKLOADS AND DEPLOYMENTS. EACH GIVES A RESULT THAT COULD SUPPORT OR REVISE THE PROPOSED EXPLANATION.

| Question and unresolved mechanism | Controlled comparison | Discriminating result |
|---|---|---|
| **Q1. Which signals survive execution changes?** Do attacks identify recurring task structure or features tied to its implementation? | Keep task labels and observation rights fixed. Hold out compositions, coordination and client/scaffold versions separately, then jointly. Compare destination, volume/timing, sequence and recipient-visible feature families under declared frozen and adapted training. | Identify a feature family whose inference survives specified changes, or a failure tied to one changed condition. Report per-axis and joint transfer with attempted-run coverage; a pooled average cannot establish invariance. |
| **Q2. How does feedback reach model traffic?** Which part is content, length or later action structure? | Fix initial instructions and task-required fields. Pair length-matched neutral padding with validity-preserving replacement of supplementary content; hold encoding and follow-up policy fixed where possible. Vary record count separately. Compare prefix, first-feedback and complete model views under tool-only and joint protection. | Additional held-out inference after feedback, with controls that exclude the competing path, supports that attribution. Report an effect interval against a prespecified meaningful-effect threshold. A nonsignificant difference alone does not rule out leakage. |
| **Q3. Can cover preserve task semantics?** When should it stop at transport or execute in isolation? | On matched original tasks, compare unprotected execution, transport-terminated cover and isolated executed cover under declared resource budgets. Refit permitted attackers; measure bytes, delay, original-task completion, quota use, state changes and feedback entering the planner. | A privacy–utility operating point satisfying prespecified state and task tolerances supports the design. Resource exhaustion, changed business state or impaired completion identifies the constraint it violates. |
| **Q4. Which inference survives task discovery?** Does cross-link association recover information hidden by waits and multiplexing? | Train on one concurrency/load and test new compositions, loads and gateways. Compare tool-only and associated tool/model views with observable flow identifiers; keep ground-truth task and stream mappings outside attacker inputs. Include real unrelated traffic and constant-output detector references. | Measure interval recall, false alarms per unit time and correctly localized inference together. Gains must survive changed workloads and loads at a stated false-alarm budget. |

Availability. The accompanying review package connects the forty-four-setting evidence matrix, transfer relations, eligibility ledger, source locators and dependency calculations to the claims. Synthetic metadata, predictions, outcome markers, uncertainty scripts and manifests provide an offline verification route. Raw-to-metadata reconstruction needs the excluded captures; the mutable remote alias cannot reproduce fixed weights. The supplement presents the current study and source basis; the artifact also preserves the earlier experimental reports and all saved measurements. Compilable LaTeX and both PDFs accompany the package.

## 11. Conclusion

Classical traffic-analysis knowledge carries into Agent security through explicit access, observation and intervention conditions. The transfer matrix identifies two execution boundaries requiring additional checks: feedback crosses separately protected services, and executed cover changes application work. The fifteen-source evidence map ties these boundaries to actual observation rights and measurements. Among four query/workflow privacy settings, adaptation and original-task utility are located in different records; the published executed-decoy result leaves matched task utility open. The controlled feedback pilot retains its inconclusive adjusted comparisons and heterogeneous complete-execution outcomes. The practical priorities follow from the synthesis: evaluate every permitted residual view, and bind privacy, resource use and original-task outcomes to the same intervention.

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